

Assignment 2: Coding Basics

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OVERVIEW

This exercise accompanies the lessons/labs in Environmental Data Analytics on coding basics.

Directions

1. Rename this file `<FirstLast>_A02_CodingBasics.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. After Knitting, submit the completed exercise (PDF file) to Canvas.

Basics, Part 1

1. Generate a sequence of numbers from one to 55, increasing by fives. Assign this sequence a name.
2. Compute the mean and median of this sequence.
3. Ask R to determine whether the mean is greater than the median.
4. Insert comments in your code to describe what you are doing.

```
#1. Used seq function to create sequence from 1 to 55 by 5s and made it an object  
one_to_55_by_5s <- seq(1,55,5)  
one_to_55_by_5s
```

```
## [1] 1 6 11 16 21 26 31 36 41 46 51
```

```
#2. Used mean and median functions to do computations on sequence  
mean(one_to_55_by_5s)
```

```
## [1] 26
```

```
median(one_to_55_by_5s)
```

```
## [1] 26
```

```
#3. Wrote a statement comparing the outputs of mean and median functions
mean(one_to_55_by_5s)>median(one_to_55_by_5s)
```

```
## [1] FALSE
```

Basics, Part 2

5. Create three vectors, each with four components, consisting of (a) student names, (b) test scores, and (c) whether they are on scholarship or not (TRUE or FALSE).
6. Label each vector with a comment on what type of vector it is.
7. Combine each of the vectors into a data frame. Assign the data frame an informative name.
8. Label the columns of your data frame with informative titles.

```
#5. & 6.
names_vector <- c("Dan","Stephanie","Linda","Marcus") #character vector
names_vector
```

```
## [1] "Dan"      "Stephanie" "Linda"      "Marcus"
```

```
scores_vector <- c(61, 48, 55, 42) #numeric/integer vector
scores_vector
```

```
## [1] 61 48 55 42
```

```
scholarship_vector <- c(TRUE,TRUE,FALSE,FALSE) #logical vector
scholarship_vector
```

```
## [1] TRUE TRUE FALSE FALSE
```

```
#7.
student_info_df <- data.frame(names_vector,scores_vector,scholarship_vector)
student_info_df
```

```
##   names_vector scores_vector scholarship_vector
## 1      Dan           61             TRUE
## 2  Stephanie           48             TRUE
## 3     Linda           55            FALSE
## 4     Marcus           42            FALSE
```

```
#8.
names(student_info_df) <- c("Student Name","Exam Score","On Scholarship?")
student_info_df
```

```
##   Student Name Exam Score On Scholarship?
## 1      Dan           61             TRUE
## 2  Stephanie           48             TRUE
## 3     Linda           55            FALSE
## 4     Marcus           42            FALSE
```

9. QUESTION: How is this data frame different from a matrix?

Answer: This data frame contains columns of different modes/vector types. A matrix must contain data of the same type.

10. Create a function with one input. In this function, use `if...else` to evaluate the value of the input: if it is greater than 50, print the word “Pass”; otherwise print the word “Fail”.
11. Create a second function that does the exact same thing as the previous one but uses `ifelse()` instead of `if...else`.
12. Run both functions using the value 52.5 as the input
13. Run both functions using the **vector** of student test scores you created as the input. (Only one will work properly...)

```
#10. Create a function using if...else
greater_than_50_if...else <- function(x) {
  if(x>50) {"Pass"}
  else {"Fail"}
}
```

```
#11. Create a function using ifelse()
greater_than_50_ifelse <- function(x) {
  ifelse(x>50,"Pass","Fail")
}
```

```
#12a. Run the first function with the value 52.5
greater_than_50_if...else(52.5)
```

```
## [1] "Pass"
```

```
#12b. Run the second function with the value 52.5
greater_than_50_ifelse(52.5)
```

```
## [1] "Pass"
```

```
#13a. Run the first function with the vector of test scores
# greater_than_50_if...else(scores_vector)
```

```
#13b. Run the second function with the vector of test scores
greater_than_50_ifelse(scores_vector)
```

```
## [1] "Pass" "Fail" "Pass" "Fail"
```

14. QUESTION: Which option of `if...else` vs. `ifelse` worked? Why? (Hint: search the web for “R vectorization”)

Answer: The ‘ifelse’ version worked. This is because ‘ifelse’ is a vectorized function, meaning that it can operate on each element of an input vector and produce a vector output of equal length. The ‘if...else’ version failed because ‘if’ is not vectorized and cannot operate on an input of more than one value.

NOTE Before knitting, you’ll need to comment out the call to the function in Q13 that does not work. (A document can’t knit if the code it contains causes an error!)